MATHEMATICS

PERMUTATION AND COMBINATION

THEORY SHEET

ENGLISH

Piprali Road, Sikar
Ph. 01572-241911, www.matrixedu.in

1.1 INTRODUCTION - FUNDAMENTAL PRINCIPLE OF COUNTING

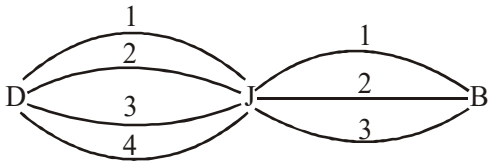
Q.1 There are four ways from Delhi to Jaipur and three ways from Jaipur to Bombay.

(i) In how many ways a person can travel from Delhi to Bombay, via Jaipur?

(ii)A person is in Jaipur now, in how many ways he can leave Jaipur ?

Ans. (i) 12 (ii) 7

Sol.



(i) $4 \times 3 = 12$ (ii) $4 + 3 = 7$

Fundamental Principle of Counting

If an event A can occur in 'm' different ways and another event B can occur in 'n' different ways, then the total number of different ways of-

(a) simultaneous occurrence of both events in a definite order is m× n. This can be extended to any number of events (known as multiplication principle).

(b) happening exactly one of the events is m + n (known as addition principle).

Q.2 A cinema hall has 5 gates(G_1, G_2, G_3, G_4 and G_5). In how many ways a person can enter the cinema hall and exit the cinema hall.

(ii) by using a different gate. (iii) if he can enter by G_1 or G_2 only.

(iv) if he can enter by G_1 or G_2 only and can exit by G_3, G_4 or G_5 only.

Ans. (i) 5×5 (ii) 5×4 (iii) 2×5 (iv) 2×3

Sol. (i) $5 \times 5 = 25$

(ii) $5 \times 4 = 20$

(iii) $2 \times 5 = 10$

(iv) $2 \times 3 = 6$

Q. 3 A boy has 4 jeans, 5 shirts and 3 caps, in how many ways he can be dressed up ?

Ans. $4 \times 5 \times 3$

Sol. No. of ways to choose jeans = 4

No. of ways to choose shirts = 5

No. of ways to choose caps = 3

Total = $4 \times 5 \times 3 = 60$

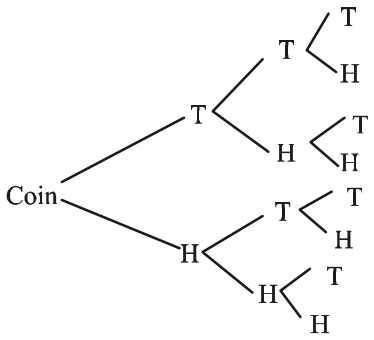
Q. 4 A king has 7 queens, each queen has 7 sons, each son has 7 wives and each wife has 7 cats, find total number of cats.

Ans. 7^4

Sol. $7 \times 7 \times 7 \times 7 = 7^4$

Q. 5 A coin is tossed three times, find total no of possible results.

Ans 8

Sol 

$(2\times2\times2)$

Q. 6 There are four books P, C, M and E which are to be placed on a shelf. Find total number of ways.

Ans 24

Sol. P|C|M|E

$= 4 \times 3 \times 2 \times 1 = 24$

Q. 7 Ten players: $p_1, p_1 \ldots\ldots\ldots\ldots p_{10}$, are participating in a swimming competition. In how may ways, first three position can be occupied.

Ans 720

Sol. 1. 10

×

2. 9

×

3. 8

= 720

Q. 8 Five chairs $C_1, C_2, C_3 \ldots\ldots.., C_5$ are to be occupied by five girls $G_1, G_2, G_3 \ldots\ldots.., G_5$. Find total number of ways.
(ii) If C_1 is to be occupied by G_1 or G_2 only. (iii) If G_1 can not sit on C_1.
(iv) If G_1 sits on C_1 or C_2 only and G_2 sits on C_2 or C_3 only.

Ans. (i) 120 (ii) 48 (iii) 96 (iv) 18

Sol. C_1 C_2 C_3 C_4 C_5

(i) $5 \times 4 \times 3 \times 2 \times 1 = 120$

(ii) $\underline{G_1, G_2}$ ___ ___ ___ ___

$2 \times 4 \times 3 \times 2 \times 1 = 48$

(iii) G_1 — C_2, C_3, C_4, C_5

$4 \times (4 \times 3 \times 2 \times 1) = 96$

(iv) G1 G2 ____ ____ ____

G1 ____ G2 ____ ____

____ G1 G2 ____ ____

$3 \times (3 \times 2 \times 1) = 18$

Q. 9 There are ten true/false questions in a question paper. A student has to attempt all the questions. Find total number

of ways

(ii) If all the answers are correct. (iii) If all the answers are wrong.

(iv) If at least one answer is correct. (v) If at least one answer is wrong.

Ans. (i) 2^{10} (ii) 1 (iii) 1 (iv) $2^{10}-1$ v) $2^{10}-1$

Note: If "atleast" is mentioned in question then think of complementary approach.

Sol. Q_1 Q_2 Q_3 ..... Q_{10}

$\frac{T}{F}$ $\frac{T}{F}$ $\frac{T}{F}$ $\frac{T}{F}$

(i) $2 \times 2 \times 2 \times \ldots\ldots 2 = 2^{10}$

(ii) $1 \times 1 \times 1 \times \ldots\ldots 1 = 1$

(iii) $1 \times 1 \times 1 \times \ldots\ldots 1 = 1$

(iv) Total – all wrong

$2^{10} - 1$

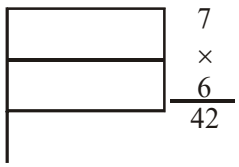
(v) Total – all correct

$2^{10} - 1$

Q. 10 There are seven different flags available. In how many ways a signal can be transmitted using two flags one above the other.

Ans 42

Sol.



$$\begin{array}{r} 7 \\ \times \\ 6 \\ \hline 42 \end{array}$$

1.2 FORMATION OF NUMBERS

Q. 1 1, 2, 3, 4, 5 are available :

(i)How many 5 digits numbers without repetition can be formed.

(ii) How many 5 digit numbers can be formed If repetition is allowed.

(iii) How many 3-digit numbers without repetition can be formed.

(iv) How many 3 digit numbers can be formed If repetition is allowed.

(v) How many 6-digit numbers without repetition can be formed.

(vi) How many 6 digit numbers can be formed If repetition is allowed.

Ans. (i) 120 (ii) 5^5 (iii) 60 (iv) 5^3 (v) 0 (vi) 5^6

Sol. (i) $5 \times 4 \times 3 \times 2 \times 1 = 120$

(ii) $5 \times 5 \times 5 \times 5 \times 5 = 5^5$

(iii) $5 \times 4 \times 3 = 60$

(iv) $5 \times 5 \times 5 = 5^3$

(v) 0

(vi) $5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^6$

1.3 DIVISIBILITY RULES

The following chart shows the conditions of divisibility of numbers by 2,3,4,5,6,8,9,11, 25

Divisible by	Condition
2	whose last digit is even (2, 4, 6, 8, 0)
3	sum of whose digits is divisible by 3
4	whose last two digits number is divisible by 4
5	whose last digit is either 0 or 5
6	which is divisible by both 2 and 3
8	whose last three digits number is divisible by 8
9	sum of whose digits is divisible by 9
11	$\begin{vmatrix}\text{sum of digits} & -\text{sum of digits}\\ \text{at odd places} & \text{at even places}\end{vmatrix}$ is divisible by 11
25	whose last two digit number is divisible by 25

Q. 1 (i) How many 3-digit numbers can be formed.

(ii) How many 3-digit numbers can be formed without repetition.

(iii) How many 3-digit numbers can be formed if all the digits are odd without repetition.

(iv) How many 3-digit even numbers can be formed if repetition is allowed.

(v) How many 3-digit even numbers can be formed without repetition.

(vi) How many 3-digit numbers can be formed containing at least one even digit .

(vii) How many 3-digit numbers can be formed such that at least one digit is 7 .

(viii) How many 4-digit numbers can be formed without repetition divisible by 5.

(ix) How many 3-digit numbers can be formed greater than 400 without repetition.

(x) How many 5-digit numbers can be formed without repetition divisible by 3 using 0, 1, 2, 3, 4, 5.

(xi) How many 4-digit numbers can be formed without repetition divisible by 4 using 0, 1, 2, 3, 4, 5.

Ans. (i) 900 (ii) 648 (iii) 60 (iv) 450 (v) 328 (vi) 775
(vii) 252 (viii) 952 (ix) 432 (x) 216 (xi) 72

Sol. (i) $9 \times 10 \times 10 = 900$

(ii) $9 \times 9 \times 8 = 648$

(iii) $5 \times 4 \times 3 = 60$

use (1,3,5,7,9) only

(iv) ____ ____ ____
↓ (0,2,4,6,8)
0 not comes

$9 \times 10 \times 5 = 450$

(v) __ __ $\underline{0} \Rightarrow 9 \times 8 \times 1 = 72$

+

__ __ __ $(2,4,6,8) \Rightarrow 8 \times 8 \times 4 = 256$

$= 328$

(vi) Total – all odd

$900 - 5 \times 5 \times 5 = 775$

(vii) Total – without 7

$900 - 8 \times 9 \times 9$

252

(viii) $\overline{9} \times \overline{8} \times \overline{7} \times \overset{0}{\overline{1}} = 504$

$_\ _\ _\ \underline{5}$ +

$8 \times 8 \times 7 \times 1 = 448$

952

(ix) $6 \times 9 \times 8 = 432$

(x) 0,1,2,3,4,5

sum = 0 + 1 + 2 + 3 + 4 + 5 = 15

$1,2,3,4,5 \Rightarrow$ sum = 15

$0,1,2,4,5 \Rightarrow$ sum = 12

$5 \times 4 \times 3 \times 2 \times 1$ ...(1)

$4 \times 4 \times 3 \times 2 \times 1$ ...(2)

Ans. = (1) + (2)

= 120 + 96 = 216

(xi) 0,1,2,3,4,5

__ __ __ __

0, 4

1, 2

2, 0

2, 4

3, 2

4, 0

5, 2

it last two digit having zero.

__ __ 0 4

__ __ 2 0

__ __ 4 0

$4 \times 3 \times 3$ ...(1)

__ __ 2 4

__ __ 1 2

__ __ 3 2

__ __ 5 2

$3 \times 3 \times 4$ ...(2)

Add = (1) + (2)

= 72

1.4 FORMATION OF WORDS

Q.1 Using letters of MOBILES, How many four letter words that can be formed?

(ii) Which starts with M. (iii) Which starts with a vowel

(iv) Which contains M. (v) Which contains all the vowels.

(vi) Which contains at least one vowel. (vii) Which starts with a vowel or ends with a vowel

Ans. (i) 840 (ii) 120 (iii) 360 (iv) 480 (v) 96

(vi) 816 (vii) 600

Sol. M O B I L E S

(i) $7 \times 6 \times 5 \times 4 = 840$

(ii) M __ __ __

$1 \times 6 \times 5 \times 4$

(iii) $3 \times (6 \times 5 \times 4)$

O, I, E

(iv) Total – without M

$7 \times 6 \times 5 \times 4 - 6 \times 5 \times 4 \times 3$

(v) O I E M $\Rightarrow 4 \times 3 \times 2 \times 1$

O I E B $\Rightarrow 4 \times 3 \times 2 \times 1$

O I E L $\Rightarrow 4 \times 3 \times 2 \times 1$

O I E S $\Rightarrow 4 \times 3 \times 2 \times 1$

$= 4 \times (4 \times 3 \times 2 \times 1)$

(vi) Total – without vowel

$7 \times 6 \times 5 \times 4 - 4 \times 3 \times 2 \times 1$

(vii) $\underline{v}$ _ _ _ $\Rightarrow 3 \times (6 \times 5 \times 4)$ ...(1)

_ _ _ $\underline{v} \Rightarrow (6 \times 5 \times 4) \times 3$ ...(2)

$\underline{v}$ _ _ $\underline{v} \Rightarrow (3 \times 2) \times (5 \times 4)$ ...(3)

Ans = (1) + (2) – (3)

$\Rightarrow 360 + 360 - 120$

$\Rightarrow 600$

1.5 m^n OR n^m

Q. 1 Six persons are to be seated in 5 railways compartments, find number of ways.

Ans. 5^6

Sol. $5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^6$

Q. 2 Three rings are to be put in four fingers find total number of ways.

Ans. 4^3

Sol. $4 \times 4 \times 4 = 4^3$

1.6 QUESTION ON RANK OF WORD

Q.1 All the words that can be formed using all the letters of RAPID are arranged in alphabetical order, then find rank of RAPID.

Ans 102

Sol. First of all, arrange all letters of given word alphabetically : 'ADIPR'

Total number of words starting with A _ _ _ _ = 4! = 24

Total number of words starting with D _ _ _ _ = 4! = 24

Total number of words starting with I _ _ _ _ = 4! = 24

Total number of words starting with P _ _ _ _ = 4! = 24

Total number of words starting with RAD _ _ = 2! = 2

Total number of words starting with RAI _ _ = 2! = 2

Total number of words starting with RAPD _ = 1

Total number of words starting with RAPI _ = 1

$\therefore$ Rank of the word RAPID = 24 + 24 + 24 + 24 + 2 + 2 + 1 + 1 = 102 **Ans.**

Q.2 QUEST.

Ans 43

Sol. Number of words beginning with E = ${}^4P_4 = 24$
Number of words beginning with QE = ${}^3P_3 = 6$
Number of words beginning with QS = 6
Number of words beginning withQT = 6.
Next word is 'QUEST'

$\therefore$ its rank is 24 + 6 + 6 + 6 + 1 = 43.

Q.3 TOUGH

Ans 89

Sol. T O U G H

G H O T U (Alphabetical Order)

G _ _ _ _ $\Rightarrow$ 24

H _ _ _ _ $\Rightarrow$ 24

O _ _ _ _ $\Rightarrow$ 24

T G _ _ _ $\Rightarrow$ 6

T H _ _ _ $\Rightarrow$ 6

T O G _ _ $\Rightarrow$ 2

T O H _ _ $\Rightarrow$ 2

TOUGH = 1

= 24 + 24 + 24 + 6 + 6 + 2 + 2 + 1 = 89

2. FACTORIAL

The continuous product of first n natural numbers is called **factorial** and it can be represented by notation $\underline{|n}$ or n! .

So n! = 1.2.3..... (n–1) .n

or n! = n (n–1) (n–2) 3.2.1

$= n \{ (n-1)(n-2) \ldots\ldots 3.2.1. \}$

$\therefore \quad n! = n(n-1)! = n(n-1)(n-2)! = n(n-1)(n-2)(n-3)!$

$n(n-1)\ldots\ldots\ldots(n-r+1) = \frac{n!}{(n-r)!}$

Some useful results :

1. $0! = 1 \quad 1! = 1 \quad 2! = 2$
   $3! = 6 \quad 4! = 24 \quad 5! = 120$
   $6! = 720 \quad 7! = 5040 \quad 8! = 40320$
2. $(-n)!$ = Meaningless
3. $(2n)! = 2^n.n! [1.3.5.7\ldots\ldots(2n-1)]$

Q.1 Find the exponent of 2 in 100!

Ans. 97

Sol. Exponent of 2 in 100!

$$= \left[\frac{100}{2}\right] + \left[\frac{100}{4}\right] + \left[\frac{100}{8}\right] + \left[\frac{100}{16}\right] + \left[\frac{100}{32}\right] + \left[\frac{100}{64}\right] + \ldots.$$

$= 50 + 25 + 12 + 6 + 3 + 1$

$= 97$

Q.2 Find the exponent of 3 in 100!

Ans. 48

Sol. Exponent of 3 in 100!

$$= \left[\frac{100}{3}\right] + \left[\frac{100}{9}\right] + \left[\frac{100}{27}\right] + \left[\frac{100}{81}\right] + 0 + 0\ldots.$$

$= 33 + 11 + 3 + 1$

$= 48$

Q.3 Find the exponent of 10 in 100!

Ans. 24

Sol. Exponent of 10 in 100!

$10 = 5 \times 2$ {We take large no.}

$$= \left[\frac{100}{5}\right] + \left[\frac{100}{25}\right] + \left[\frac{100}{125}\right] + \ldots..$$

$= 20 + 4 = 24$

Q.4 Find the exponent of 6 in 50!

Ans 22

Sol. $E_2(50!)=\left[\frac{50}{2}\right]+\left[\frac{50}{4}\right]+\left[\frac{50}{8}\right]+\left[\frac{50}{16}\right]+\left[\frac{50}{32}\right]+\left[\frac{50}{64}\right]$ (where [ ] denotes integral part)

$E_2(50!) = 25 + 12 + 6 + 3 + 1 + 0 = 47$

$E_3(50!)=\left[\frac{50}{3}\right]+\left[\frac{50}{9}\right]+\left[\frac{50}{27}\right]+\left[\frac{50}{81}\right]$

$E_3(50!) = 16 + 5 + 1 + 0 = 22$

Q.5 If $\sum_{r=1}^{10} r!\left(r^3+6r^2+2r+5\right)=\alpha(11!)$, , then the value of α is equal to _____. **[O]**

यदि $\sum_{r=1}^{10} r!\left(r^3+6r^2+2r+5\right)=\alpha(11!)$ है, तो α का मान बराबर है _____ **[O]**

Ans. 160

Sol. $\sum_{r=1}^{10} \underline{|r}\left(r^3+6r^2+2r+5\right)=\alpha\underline{|11}$

$\sum_{r=1}^{10} \underline{|r}\left((r+1)(r+2)(r+3)-9(r+1)+8\right)$

$\sum_{r=1}^{10} \underline{|r+3}-9\underline{|r+1}+8\underline{|r}$

$\sum_{r=1}^{10} \underline{|r+3}-\underline{|r+1}-8\underline{|r+1}+8\underline{|r}$

$\sum_{r=1}^{10} \underline{|r+3}-\underline{|r+1}-8\sum_{r=1}^{10}\underline{|r+1}-\underline{|r}$

$=160\underline{|11}=\alpha\underline{|11}$

$\alpha = 160$

Note:

Prime factorisation of n! : Let p be a prime number and n be a positive integer, then exponent of p in n! is denoted by E_p (n!) and is given by

$E_p(n!) = \left[\frac{n}{p}\right] + \left[\frac{n}{p^2}\right] + \left[\frac{n}{p^3}\right] + ... + \left[\frac{n}{p^k}\right]$

where $p^k \le n < p^{k+1}$ and [x] denotes the integral part of x.

3.1 DEFINITION OF PERMUTATION & COMBINATION

Permutation : Each of the arrangements in a definite order which can be made by taking some or all of the things at a time is called a PERMUTATION.

Combination : Each of the groups or selections which can be made by taking some or all of the things without

considering the order of the things in each group is called a COMBINATION.

In permutation order of things matter while in selection order does not matters. for eq. ABC & ACB are different permutations but same combmbinations.

Useful Theorems

1. Number of permutations of n **different** things, taken r at a time = ${}^nP_r$

$${}^nP_r = n(n-1)(n-2) \ldots\ldots\ldots\ldots (n-r+1) = \frac{n!}{(n-r)!}$$

Note :

(i) ${}^nP_n = n!$, ${}^nP_0 = 1$, ${}^nP_1 = n$

(ii) Number of arrangements of n **distinct** things taken all at a time = n!

(iii) ${}^nP_r$ is also denoted by A_r^n or P(n,r).

2. ${}^nC_r$ denotes the number of combinations of n different things taken r at a time = ${}^nC_r$

$${}^nC_r = \frac{n!}{r!(n-r)!} = \binom{n}{r} = C(n, r).$$

Q.1 In how many ways can a committee of 6 persons be made out of 10 persons ?

Ans 210

Sol. We have to select 6 persons from 10 given persons. This can be done in ${}^{10}C_6$ ways therefore number of committees.

$$= {}^{10}C_6 = {}^{10}C_4 = \frac{10!}{4!\,6!} = \frac{10.9.8.7}{4.3.2.1} = 210$$

3.2 PROPERTIES OF ${}^nC_r$

(i) ${}^nP_r = {}^nC_r \cdot r!$ (ii) ${}^nC_r = {}^nC_{n-r}$; ${}^nC_0 = {}^nC_n = 1$

(iii) ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$ (iv) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(v) ${}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1}$ (vi) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n+1-r}{r}$

(viii) ${}^nC_0 = 1$; ${}^nC_1 = n$; ${}^nC_2 = \frac{n(n-1)}{2!}$; ${}^nC_3 = \frac{n(n-1)(n-2)}{3!}$

3.3 QUESTIONS ON SELECTION AND ARRANGEMENTS

Q. 1 There are 50 boys and 40 girls in a class. A team of 11 players is to be selected find total no of ways.

(ii) If all of them are boys. (iii) If all of them are girls.

(iv) If there are 6 boys and 5 girls in the team. (v) If there is at least one girl.

(vi) If there is at most one girl.

Ans. (i) ${}^{90}C_{11}$ (ii) ${}^{50}C_{11}$ (iii) ${}^{40}C_{11}$ (iv) ${}^{50}C_6\,{}^{40}C_5$

(v) ${}^{90}C_{11} - {}^{50}C_{11}$ (vi) ${}^{40}C_1\,{}^{50}C_{10} + {}^{50}C_{11}$

Q. 2 From a pack of 52 cards we have to select 4 cards. find total no of ways

(ii) If all of them are face cards (iii) If all of them are black (iv) If all of them are jack

(v) If all of them are hearts (vi) If there are exactly two kings (vii) If all of them are of same colour.
(viii) If all of them are of same suit (ix) If there is at least one king (x) If there is at most one king

Ans. (i) $^{52}C_4$ (ii) $^{12}C_4$ (iii) $^{26}C_4$ (iv) $^{4}C_4$
(v) $^{13}C_4$ (vi) $^{48}C_2\ ^{4}C_2$ (vii) $^{2}C_1{}^{26}C_4$ (viii) $^{4}C_1\ ^{13}C_4$
(ix) $^{52}C_4 - {}^{48}C_4$ (x) $^{48}C_4 + {}^{48}C_3\ ^{4}C_1$

Q. 3 There are 10 different consonants and 4 different vowels available. How many 5 letter words that can be formed which contain 3 consonants and 2 vowels
(ii)Such that terminal positions are occupied by vowels only

Ans. (i) $^{10}C_3\ ^{4}C_2\ (5!)$ (ii) $^{10}C_3\ ^{4}C_2\ (2!)\ (3!)$

Sol. (i) $\begin{matrix}10C\\4V\end{matrix} \rightarrow 3C\,2V$

$(^{10}C_3 \cdot {}^{4}C_2) \times 5!$

(ii) V _ _ _ V

$(^{10}C_3 \cdot {}^{4}C_2) \times (2! \times 3!)$

Q. 4 $P_1, P_2 \ldots\ldots\ldots P_{10}$. In how many ways we can select 5 persons if P_1 and P_2 always come together (if they come)

Ans. $^{8}C_3\ ^{2}C_2 + {}^{8}C_5$

Sol. $P_1, P_2, \ldots\ldots P_{10}$

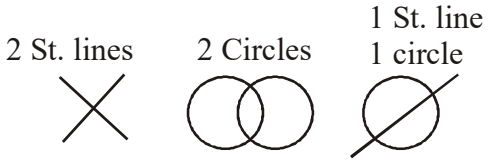
$\Rightarrow P_1P_2$ Included + P_1P_2 Excluded

$^{2}C_2 \cdot {}^{8}C_3 + {}^{8}C_5$

Q. 5 There are eight different straight lines and six different circles in a plane. Find maximum number of points of intersection

Ans. 154

Sol.

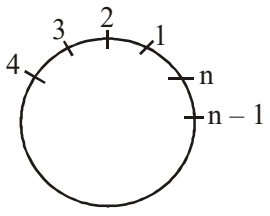


$^{8}C_2 \times 1 + {}^{6}C_2\ 2 + {}^{8}C_1\ {}^{6}C_1 \times 2 = 154$

Q. 6 (i) There are n points in a plane such that no three of which are collinear. Find number of straight lines that can be formed by joining two of them.
(ii) How many triangles that can be formed by joining them.

Ans. (i) $^{n}C_2$ (ii) $^{n}C_3$

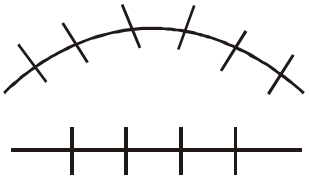
Sol. (i)



(i) $^{n}C_2$
(ii) $^{n}C_3$

Q.7 10 points in a plane no three being collinear except 4 which are in the same line.

Find (1) no. of lines ; (2) no. of triangles

Sol 

No. of lines = ${}^{10}C_2 - (\text{no of extra lines})$

no. of extra lines = ${}^4C_2 - 1$

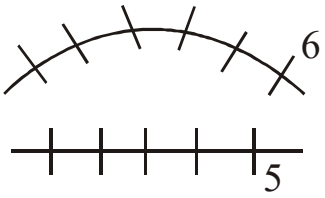
no. of lines = ${}^{10}C_2 - (4_{C_2} - 1)$

no. of Δ = ${}^{10}C_3 - (4_{C_3} - 0)$

M–II

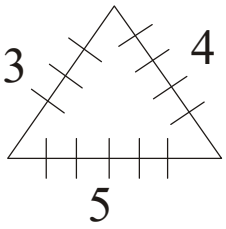
no. of lines = ${}^bC_2 + {}^6C_1 {}^4C_1 + 1$

no. of $\Delta = {}^6C_3 + {}^6C_2 {}^4C_1 + {}^6C_1 {}^4C_2 + 0$

Q.8 

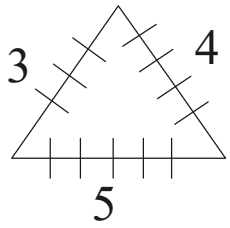
Ans no. of lines = ${}^{11}C_2 - \left({}^5C_2 - 1\right)$; no. of $\Delta = {}^{11}C_3 - ({}^5C_3 - 0)$

Sol. (i) ${}^{11}C_2 - {}^5C_2 + 1$
(ii) ${}^{11}C_3 - {}^5C_3$

Q.9 

Vertices of triangles are also included

Ans no. of lines = ${}^{15}C_2 - ({}^5C_2 - 1) - ({}^6C_2 - 1) - ({}^7C_2 - 1)$; no. of $\Delta = {}^{15}C_3 - ({}^5C_3 - 0) - ({}^6C_3 - 0) - ({}^7C_3 - 0)$

Sol. 

3 Vertices

(i) $^{15}C_2 - (^7C_2 + ^6C_2 + ^5C_2) + 3$

(ii) $^{15}C_3 - (^7C_3 + ^6C_3 + ^5C_3)$

Q.10 Find number of diagonals in a n sided polygon .

Ans. $^nC_2 - n$

Sol. $^nC_2 - n$

Total sides

Q.11 Find total number of 5 digit numbers N = abcde if

(i) a > b > c >d > e (ii) a < b < c < d < e

Ans. (i) $^{10}C_5$ (ii) $^9C_5$

Sol. N = abcde

(i) a > b > c > d > e

$^{10}C_5 \times 1$

(ii) a < b < c < d < e

So, $^9C_5 \times 1$

Q.12 Find total number of selections of 4 persons from 6 married couples such that

(i) They form no couple (ii) They form exactly 1 couple

(iii)They form exactly two couples

Ans. (i) 240 (ii) 240 (iii) 15

Sol. M_1 W_1

M_2 W_2

M_3 W_3

M_4 W_4

M_5 W_5

M_6 W_6

(i) 4M + 3M 1W + 2M 2W + 1M 3W + 4W

$^6C_4 + ^6C_3 \times ^3C_1 + ^6C_2\ ^4C_2 + ^6C_1\ ^5C_3 + ^6C_4$

or

Select 4 couple

$^6C_4 \cdot (^2C_1 \cdot ^2C_1 \cdot ^2C_1 \cdot ^2C_1)$

Select 1 from all 4 couple

(ii) M_1W_1 M_2 M_3

$^6C_1 \cdot \{^5C_2 \cdot (^2C_1 \cdot ^2C_1)\}$

↓ ↓

One couple One each from other two couples

(iii) $^6C_2$

Q.13 {1, 2, 3, ... 51} In how many ways we can select 3 numbers which are in A.P.

Ans $^{25}C_2 + {}^{26}C_2$

Sol $\{1, 2, 3, ..., 51\}$

$a, b, c \Rightarrow 2b = a + c$

$$b = \left(\frac{a+c}{2}\right) \Rightarrow a + c \in \text{Even}$$

Both even + Both odd

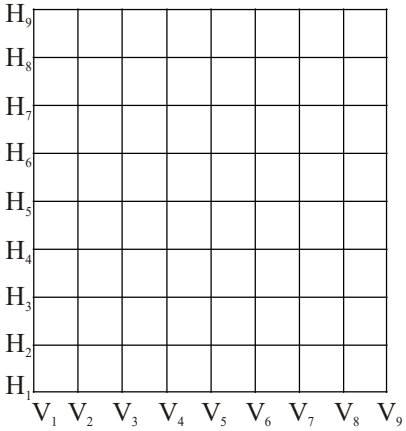
$\Rightarrow {}^{25}C_2 + {}^{26}C_2$

Q.14 In a chess board of 9 horizontal lines and 9 vertical lines, find total

(i) Number of rectangles (ii) Number of squares

Ans (i) 1296 (ii) 204

Sol



(i) no. of rectangles $^{9}C_2 \times {}^{9}C_2$

(ii) no. of squares

side = 1	8×8
side = 2	7×7
side = 3	6×6
side = 4	5×5
side = 5	4×4
side = 6	3×3
side = 7	2×2
side = 8	1×1

Q.15 In a board of 6 horizontal lines and 7 vertical lines, find

(i) Number of rectangles (ii) Number of squares

Ans (i) 315 (ii) 70

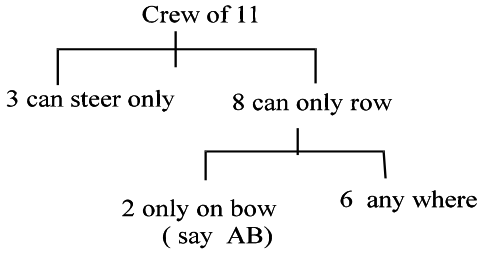
Sol (i) no. of rectangles $= {}^{7}C_2 \times {}^{6}C_2$

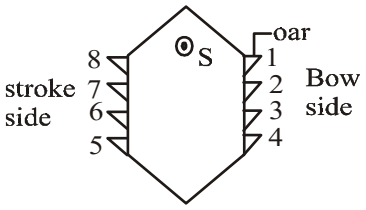
(ii) No. of squares = 6×5
5×4
4×3
3×2
2×1

Q. 16 From a crew of 11 people, 3 can steer but cannot row and remaining 8 can row but cannot steer. Out of the 8 persons who can row, 2 can row only on bow side. In how many ways staff can be arranged on the boat ?

Ans 25920

Sol





– position for steering can be selected in $^3C_1$ ways
– Two positions on the bow side can be selected in $^4C_2$ ways and AB can be arranged in 2! ways .
– Remaining 6 people on 6 positions can be seated in 6! ways
Hence using FPC Total ways $^3C_1 . {}^4C_2 .2! . 6! = 25920$ Ans]

[Alternatively: 2 out of 6 people can be selected in $^6C_2$ ways. Now ($^6C_2$ + AB)
i.e. 4 persons on bow side can be seated in 4! ways .
Total ways = $^3C_1 . {}^6C_2 . 4! . 4! = 25920$ Ans]

3.4 QUESTION ON GAP AND STRING METHOD

Q. 1 Seven different colors of a rainbow are to be arranged in a line. Find total number of ways.
(ii)If red and yellow are always together. (iii) If R, Y and O are always together.
(iv) If B and G are always together and Y and O are always together
(v) If R, Y and O are always separated. (vi) If B and G are never together.

Ans. (i) 7! (ii) (6!)(2!) (iii) (5!)(3!) (iv) (5!)(2!)(2!)
(v) (4!)($^5C_3$)(3!) (vi) (5!)($^6C_2$)(2!)

Sol. (i) 7!

(ii) $\boxed{R / Y} \rightarrow 6! \times 2!$

(iii) $\boxed{YRO} \rightarrow 5! \times 3!$

(iv) $\boxed{BG}\boxed{YO} \rightarrow 5! \times 2! \times 2!$

(v) |V|I|B|G|
$4! \times {}^5C_3 \times 3!$

(vi) $7! - 6! \times 2!$
|V| I |Y| O |R|
$5! \times {}^6C_2 \times 2!$

(vii) $|B_2|G_2|B_3|G_3|B_4|G_4|$

$(2 \times 3! \times 3!) \times {}^7C_1 \times 1$

Boys and ↓ ↓

girls alternate for B_1G_1 arrangement of B_1G_1

Q. 2 Four boys B_1, B_2, B_3, B_4 and four girls G_1, G_2, G_3, G_4 are to be seated in a line. Find total number of ways.

(ii)If all the girls are together. (iii) No two girls are together.

(iv) G_1 and G_2 are together. (v) B_1 and G_1 are not together.

(vi) Boys and girls are alternate. (vii) Boys and girls are alternate and B_1 and G_1 are together.

Ans. (i) 8! (ii) 5! 4! (iii) $(4!)({}^5C_4)(4!)$ (iv) 7! 2!

(v) $(6!)({}^7C_2)(2!)$ (vi) (2)(4!)(4!) (vii) $(3!\ 3!)(2)({}^7C_1 \times 1)$

Sol. (i) 8!

(ii) $G_1G_2G_3G_4 \rightarrow 5! \times 4!$

(iii) $|B_1|B_2|B_3|B_4|$

$4! \times {}^5C_4 \times 4!$

(iv) $G_1G_2 \rightarrow 7! \times 2!$

(v) Total – B_1G_1

$8! - 7! \times 2!$

(vi) BG BG BG BG $\rightarrow 4! \times 4!$

GB GB GB GB $\rightarrow 4! \times 4!$

$2 \times 4! \times 4!$

Q. 3 If 4 married couples are to be seated in a line then the find number of ways if each couple is together.

Ans. $(4!)\ (2!)^4$

Sol. $\boxed{M_1W_1}\boxed{M_2W_2}\boxed{M_3W_3}\boxed{M_4W_4}$

$4! \times 2! \times 2! \times 2! \times 2!$

$= 4! \times (2!)^4$

Q. 4 A cricket team comprising of A,B,C,.....J, K is to be sent for batting. If 'A' wants to bat before 'J' and 'J' wants to bat after 'G'. Then the number of batting orders if other players could go in any order.

Ans. $({}^{11}C_3 \times 2)\ (8!)$

Sol. For AJG

A	G
G	A
J	J

↓

$({}^{11}C_3 \times 2) \times 8!$

3.5 STATION METHOD

Q.1 There are 10 points in a row. In how many ways can 4 points be selected such that no two of them are consecutive ?

Ans ${}^7C_4$

Sol. Total number of remaining non-selected points = 6

. . . . . .

Total number of gaps made by these 6 points = 6 + 1 = 7

If we select 4 gaps out of these 7 gaps and put 4 points in selected gaps then the new points will represent 4 points such that no two of them are consecutive.

x . . x . x . . x .

Total number of ways of selecting 4 gaps out of 7 gaps = $^{7}C_{4}$ **Ans.**

In general, total number of ways of selection of r points out of n points in a row such that no two of them are consecutive : $^{n-r+1}C_{r}$

Q.2 10 persons are sitting in a row. In how many ways we can select three of them if adjacent persons are not selected?

Ans $^{8}C_{3}$

Sol. Let $P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9, P_{10}$ be the persons sitting in this order.
If three are selected (non consecutive) then 7 are left out.
Let P,P,P,P,P,P,P be the left out & q, q, q be the selected. The number of ways in which these 3 q's can be placed into the 8 positions between the P's (including extremes) is the number ways of required selection.
Thus number of ways = $^{8}C_{3} = 56$.

Q.3 Train having 12 station inroute has to be stopped at 5 stations such that no two stopping stations are consecutive. Find number of ways.

Alternate Language : - 12 persons are sitting in a line. In how many ways 5 persons can be selected such that no two of them are consecutive.

Ans. $^{8}C_{5}$

Q.4 20 persons are sitting in a line. In how many ways 8 persons can be selected such that no two of them are consecutive.

Ans. $^{13}C_{8}$

4.1 FORMATION OF GROUPS

(a) (i) The number of ways in which (m + n) different things can be divided into two groups such that one of them contains m things and other has n things, is $\frac{(m+n)!}{m!\ n!}$ $(m \neq n)$.

(ii) If m = n, it means the groups are equal & in this case the number of divisions is $\frac{(2n)!}{n!\ n!\ 2!}$. As in any one way it is possible to interchange the two groups without obtaining a new distribution.

(iii) If 2n things are to be divided equally between two persons then the number of ways : $\frac{(2n)!}{n!\ n!\ (2!)} \times 2!$.

(b) (i) Number of ways in which (m + n + p) different things can be divided into three groups containing m, n & p things respectively is : $\frac{(m+n+p)!}{m!\ n!\ p!}$, $m \neq n \neq p$.

(ii) If m = n = p then the number of groups = $\frac{(3n)!}{n!\ n!\ n!\ 3!}$.

(iii) If 3n things are to be divided equally among three people then the number of ways in which it can be done is $\frac{(3n)!}{(n!)^3}$.

(c) In general, the number of ways of dividing n distinct objects into ℓ groups containing p objects each and m

groups containing q objects each is equal to $\frac{n!(\ell+m)!}{(p!)^{\ell}(q!)^{m}\,\ell!m!}$

Here $\ell p + mq = n$

Ex.1 To understand the article consider 10 children to be divided into three unequal groups of 2,3 and 5.

(i) First make two groups of 2 and 8 and this can be done in $\frac{10!}{2!8!}$ way say. AB/CDEFGHIJ

Consider our such group of 8 which can be divided into two groups of 3 and 5 in $\frac{8!}{3!5!}$ ways. Hence total $= \frac{10!}{2!8!}\cdot\frac{8!}{3!5!} = \frac{10!}{2!3!5!}$

Similar explanation will be valid if initial groups in 3 and 7 and then split 7 in 2 and 5.

Ex.2 However if 10 is divided into two groups of 5 each initially, which can be done in $\frac{10!}{5!5!2!}$ ways ....(1)

One such grouping is say

A B C D E F G H I J

Consider F G H I J keeping A B C D E as it is. Now the group F G H I J can be divided into two groups of 2 and 3 is $\frac{5!}{2!3!}$ and similarly when F G H I J is kept as it is, A B C D E can be divided into two groups of 2 and 3 in $\frac{5!}{2!3!}$ ways. Hence one group (each of 5) given by (1) generated 2. $\frac{5!}{2!3!}$ different groups of 2, 3, 5.

$\therefore$ Total no. of groups $= \frac{10!}{5!5!2!}\cdot\frac{2.5!}{2!3!} = \frac{10!}{2!3!5!}$

Ex.3 Consider $T_1\,T_2\,T_3\,T_4\,T_5\,T_6$ to be divided into 3 equal groups.

When we say ${}^6C_2\,.\,{}^4C_2 = \frac{6!}{2!2!2!}$ is totally wrong why?

Selected in ${}^6C_2$	Selected in ${}^4C_2$	Rejected in ${}^4C_2$
T_1T_2	T_3T_4	T_5T_6
T_1T_2	T_5T_6	T_3T_4
T_3T_4	T_1T_2	T_5T_6
T_3T_4	T_5T_6	T_1T_2
T_5T_6	T_1T_2	T_3T_4
T_5T_6	T_3T_4	T_1T_2

R S G

Note that all these six groups are counted in (A), however they are identical. Hence the answer in (A) is as many numbers of times more more as many number of times these equal groups can be arranged i.e. 3! times. Hence the correct number of groups is equal to $\frac{6!}{2!2!2!3!}$.

In case these 6 toys are to be distributed equally between R/S/G then our answer will be = $\frac{6!\times 3!}{2!2!2!3!}$

Ex.4 Note that number of ways in which 30 Jawans can be divided into 3 equal groups = $\frac{30!}{10!10!10!}\frac{1}{3!}$
and the number of ways in which 30 Jawans can be deputed equally on 3 borders, (10 on each border) is $\frac{30!}{10!10!10!3!}$ 3!

Remember that you have to divide by as many factorials as the number of equal groups.

Ex.5 number of ways in which 200 people can be divided into 100 couples is
(100 equal groups)

$$= \frac{(200)!}{(2!)^{100}.100!} \qquad .....(B)$$

Answer in (B) can also be written as

(i) $\frac{2^{100}.100!\ (1.3.4.....199)}{2^{100}.(100)!} = 1.3.5 ..... 199.$

(ii) $\frac{(100)!(101)(102).....(200)}{2^{100}.100!} = \left(\frac{101}{2}\right).\left(\frac{102}{2}\right)....\left(\frac{200}{2}\right)$

Questions

Q. 1 4 different books are to be divided among Ram and Shyam such that they get 1 and 3 books. Find total number of ways

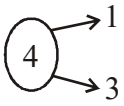
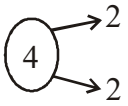
Ans. $\frac{4!}{1!3!}\times(2!)$

Q. 2 In how many ways 6 different books can be distributed among Ram, Shyam and Ghanshyam such that.
(i) They get 1 , 2, 3 books respectively.
(ii) Such that each of them gets two books.

Ans. (i) $\frac{6!}{1!2!3!}\times(1)$ (ii) $\frac{6!}{(2!)^3 3!}\times(3!)$

Q. 3 4 different books are to be distributed among Ram and Shyam such that each student gets at least one book. Find total number of ways

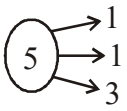
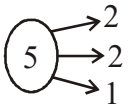
Ans. $\left(\frac{4!}{1!3!}+\frac{4!}{2!2!2!}\right)\times(2!)$

Sol.  or 

$$\frac{4!}{3!\times 1!}\times 2! \quad + \quad \left(\frac{4!}{(2!)^2}\times\frac{1}{2!}\right)\times 2!$$

Q. 4 5 different books are to be distributed among Ram, Shyam and Ghanshyam such that each student gets at least one book. Find total number of ways.

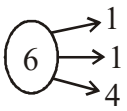
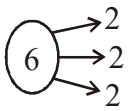
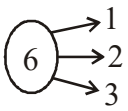
Ans. $\left(\frac{5!}{1!1!3!2!}+\frac{5!}{1!(2!)^2 2!}\right)\times 3!$

Sol.  

$$\left(\frac{5!}{3!\times 1!\times 1!}\times\frac{1}{2!}\right)+\left(\frac{5!}{2!\times 2!\times 1!}\times\frac{1}{2!}\right)\times 3!$$

Q. 5 6 different books are to be distributed among Ram, Shyam and Ghanshyam such that each student gets at least one book. Find total number of ways.

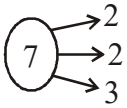
Ans. $\left(\frac{6!}{(1!)^2 4!2!}+\frac{6}{1!2!3!}+\frac{6}{(2!)^3(3!)}\right)\times(3!)$

Sol.   

$$\left(\frac{6!}{4!\times 1!\times 1!}\times\frac{1}{2!}\right)\times 3!+\left(\frac{6!}{2!\times 2!\times 2!}\times\frac{1}{3!}\right)\times 3!+\left(\frac{6!}{1!\times 2!\times 3!}\right)\times 3!$$

Q. 6 7 books are to be distributed among Ram, Shyam and Ghanshyam such that each student gets at least 2 books. Find total number of ways.

Ans. $\frac{7!}{2!2!3!2!}\times(3!)$

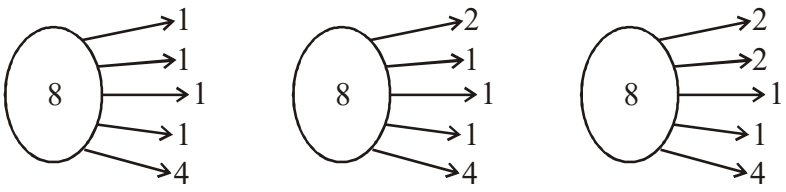
Sol. 

$$\left(\frac{7!}{2!2!3!}\times\frac{1}{2!}\right)\times 3!$$

Q. 7 8 different computers are to distributed among 5 schools such that each school gets at least one computer. Find number of ways.

Ans. $\left(\frac{8!}{(1!)^4 4!4!}+\frac{8!}{(1!)^3 2!3!3!}+\frac{8!}{(1!)^2(2!)^3 2!3!}\right)\times(5!)$

Sol.

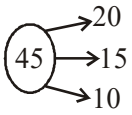


$$\left(\frac{8!}{4!}\right)5!\times\frac{1}{4!}+\left(\frac{8!}{2!\times 3!}\times\frac{1}{3!}\right)\times 5!+\left(\frac{8!}{(2!)^3\times(1!)^2}\times\frac{1}{3!}\times\frac{1}{2!}\right)\times 5!$$

Q. 8 During election 3 districts are to be convassed by 20, 15 and 10 people respectively. If 45 volunteers then the number of ways in which they can be sent.

Ans. $\frac{45!}{20!15!10!}\times 1$

Sol.



$$\frac{45!}{20!15!10!}$$

Q. 9 In how many ways 13 cards to each of the four players be distributed from a pack of 52 cards so that each may have

(i) A | K | Q | J of the same suit ; (ii) A | K | Q | J of any suit

Ans. (i) $\left(\frac{36!}{(9!)^4 4!}\times 4!\right)\times 4!$ (ii) $\left(\frac{36!}{(9!)^4 4!}\times(4!)^4\right)\times 4!$

Sol. (i)



$$4!\times\left(\frac{3!}{9!9!9!9!}\times\frac{1}{4!}\right)\times 4!$$

(ii) A A A A $\rightarrow$ 4!
K K K K $\rightarrow$ 4!
Q Q Q Q $\rightarrow$ 4!
J J J J $\rightarrow$ 4!

$$(4!)^4\times\left(\frac{36!}{(9!)^4}\times\frac{1}{4!}\right)4!$$

When to apply grouping :

When distinct things are distributed among persons such that no one remains empty.

4.2 IMPORTANT QUESTION ON FORMATION OF GROUPS

Q. 1 'n' different things are to be distributed among Ram and Shyam, such that each student gets at least one thing. Find total number of ways.

Ans $2^n - 2$

Sol $2^n - 2$ (Total; either of them doesn't gets any thing)

Q. 2 'n' different things are to be distributed among Ram, Shyam and Ghanshyam, such that each student gets at least one thing. Find total number of ways.

Ans $3^n - {}^3C_2(2^n - 2) - ({}^3C_1(1))$

Sol $= 3^n -$ (exactly one remains empty) – (exactly 2 remains empty)

$= 3^n - {}^3C_2(2^n - 2) - ({}^3C_1(1))$

Q.3 The students $S_1, S_2, ..., S_{10}$ are to be divided into 3 groups A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is _____ **[O]**

Ans. 31650

Sol. Case – I Group C has 1 student

number of ways = ${}^{10}C_1 (2^9 - 2) = A_1 = 5100$

Case – II Group C has 2 student

number of ways = ${}^{10}C_2 (2^8 - 2) = A_2 = 11430$

Case – III Group C has 3 student

number of ways = ${}^{10}C_3 (2^7 - 2) = A_3 = 15120$

Ans. $= A_1 + A_2 + A_3 = 31650$

5.1 PERMUTATION OF ALIKE OBJECTS TAKEN ALL AT A TIME

DADDY $\frac{|5}{|3}$; PAPA $\frac{|4}{|2|2}$; CLOCK $\frac{|5}{|2}$; HONOLULU $\frac{|8}{|2|2|2}$

Q.1 21 white balls and 19 black balls are to be arranged in a line. Find total number of ways.

(i) If balls of same color are different. (ii) If balls of same color are alike.

(iii) If white balls are different and black balls are alike. (iv) If white balls are alike and black balls are different.

Ans. (i) 40! (ii) $\frac{40!}{21!19!}$ (iii) $\frac{40!}{19!}$ (iv) $\frac{40!}{21!}$

Q.2 21 white balls and 19 black balls are to be arranged in a line such that no two black balls are together. Find total number of ways.

(i) If balls of same color are different. (ii) If balls of same color are alike.

(iii) If white balls are different and black balls are alike.

(iv) If white balls are alike and black balls are different.

Ans. (i) $(21!)\,{}^{22}C_{19}(19!)$ (ii) $\left(\frac{21!}{21!}\right){}^{22}C_{19}\left(\frac{19!}{19!}\right)$

(iii) $(21!)\,{}^{22}C_{19}\left(\frac{19!}{19!}\right)$ (iv) $\left(\frac{21!}{21!}\right){}^{22}C_{19}(19!)$

Q.3 How many 13 letter words that can be formed using all the letters of word ASSASSINATION

(ii) How many of them starts with A (iii) Starts with S.

(iv) Starts with S and ends with S. (v)All the vowels are together.

(vi) All the consonants are together (vii) No two vowels are together

(viii) No two consonants are together (ix) All S are together

(x)No two S are together (xi) Contains the word NATION

(xii)Without changing order of vowels. (xiii) Without changing order of consonants

(xiv) Keeping position of all vowels fixed (xv) Keeping position of each consonant fixed.

(xvi) Without changing relative order of vowels and consonants.

(xvii) No two S are together and no two N are together.

Ans. (i) $\frac{13!}{4!3!2!2!}$ (ii) $\frac{12!}{4!2!2!2!}$ (iii) $\frac{12!}{3!3!2!2!}$ (iv) $\frac{11!}{3!2!2!2!}$

(v) $\frac{8!}{4!2!}\times\frac{6!}{3!2!}$ (vi) $\frac{7!}{3!2!}\times\frac{7!}{4!2!}$ (vii) $\frac{7!}{4!2!}\times{}^{8}C_{6}\times\frac{6!}{3!2!}$ (viii) $\frac{6!}{3!2!}\times{}^{7}C_{7}\times\frac{7!}{4!2!}$

(ix) $\frac{10!}{3!2!2!}\times\frac{4!}{4!}$ (x) $\frac{9!}{3!2!2!}\times\left({}^{10}C_{4}\right)\times\frac{4!}{4!}$

(xi) $\frac{8!}{2!4!}\times1$ (xii) ${}^{13}C_{6}\times1\times\frac{7!}{4!2!}$ (xiii) ${}^{13}C_{7}\times1\times\frac{6!}{3!2!}$ (xiv) $\frac{7!}{4!2!}$

(xv) $\frac{6!}{3!2!}$ (xvi) $\frac{7!}{4!2!}\times\frac{6!}{3!2!}$ (xvii) $\left(\frac{9!}{3!2!2!}\times{}^{10}C_{4}\times\frac{4!}{4!}\right)-\left(\frac{8!}{3!2!}\times\frac{2!}{2!}\times{}^{9}C_{4}\times\frac{4!}{4!}\right)$

Sol. (i) $\frac{13!}{4!3!2!2!}$

(ii) $\underline{A}_____$

$1\times\frac{12!}{4!3!2!2!}$

(iii) $\underline{S}_____$

$1\times\frac{12!}{3!\times3!\times2!\times2!}$

(iv) $\underline{S}_____\underline{S}$

$\frac{11!}{2!3!2!2!}$

(v) $\boxed{AAAIIO}$

$$\frac{8!}{4!2!}\times\frac{6!}{2!}$$

(vi) $\boxed{\text{SSSSNTN}}$

$$\frac{7!}{3!2!}\times\frac{7!}{4!2!}$$

(vii) |S|S|S|S|N|N|T|

$$\frac{7!}{4!2!}\times {}^{8}C_{6}\times\frac{6!}{3!2!}$$

(viii) Consonants - $\underbrace{\text{SSSSNTN}}_{7}$

Vowels - |A|A|A|I|I|O|

$$\frac{6!}{3!2!}\times {}^{7}C_{7}\times\frac{7!}{4!2!}$$

(ix) $\boxed{\text{SSSS}}$

$$\frac{10!}{3!2!2!}$$

(x) |A| A |I| N |A| T |I| O |N|

$$\frac{9!}{2!3!2!}\times {}^{10}C_{4}\times 1$$

(xi) N A T I O N

$$\frac{8!}{4!2!}\times\frac{6!}{2!}$$

(xii) A A I A I O

$${}^{13}C_{6}\times 1\times\frac{7!}{4!2!}$$

(xiii) S S S S N T N

$${}^{13}C_{6}\times 1\times\frac{6!}{3!2!}$$

(xiv) $\underline{\text{A}}$ _ _ $\underline{\text{A}}$ _ _ $\underline{\text{I}}$ _ $\underline{\text{A}}$ _ $\underline{\text{I}}$ $\underline{\text{O}}$ _

$$\frac{7!}{4!2!}$$

(xv) _ S S _ S S _ N _ T _ _ N

$$\frac{6!}{3!2!}$$

(xvi) V C C V C C V C V C V V C

$$\frac{6!}{3!2!}\times\frac{7!}{4!2!}$$

(xvii) |N| N |T| A |A| A |I| I |O| – |T| A |A| A |I| I |O|

No two S togetgher – $\boxed{NN}$ and no two S together

$$\frac{9!}{2!3!2!}\times {}^{10}C_4\times 1-\frac{8!}{3!2!}\times {}^{9}C_4\times 1$$

Q.4 Find number of 11 digit numbers that can be formed using 0, 0, 0, 1, 1, 2, 3, 3, 3, 4, 4
(ii)How many of them are odd

Ans (i) ${}^{10}C_3\times\frac{3!}{3!}\times\frac{8!}{2!3!2!}$ (ii) $\left({}^{9}C_3\times\frac{3!}{3!}\times\frac{7!}{3!2!}\right)+\left({}^{9}C_3\times\frac{3!}{3!}\times\frac{7!}{2!2!2!}\right)$

Sol. 0,0,0,1,1,2,3,3,3,4,4

(i) $\frac{11!}{3!2!3!21}-1\times\frac{10!}{2!2!3!2!}$ (Total – starting with zero)

(ii) (Ending with 1) ${}^{9}C_3\times 1\times\frac{7!}{3!2!}$

$+$

(Endign with 3) ${}^{9}C_3\times 1\times\frac{7!}{3!2!\times 2!}$

Q.5 All the words that can be formed using all the letters of MADAM are arranged in alphabetical order, then find rank of MADAM.

Ans 21

Sol. M A D A M

$\underline{A}____\to\frac{4!}{2!}=12$

$\underline{D}____\to\frac{4!}{2!2!}$

$\underline{M}\,\underline{A}\,\underline{A}__\to 2!$

$\underline{M}\,\underline{A}\,\underline{D}\,\underline{A}\,\underline{M}$

Q.6 Find total number of 7 digit number whose (i) Sum of digits is 61. (ii)Sum of digits is 60.

Ans (i) 28 (ii) 84

Sol. 9 9 9 9 9 9 9

Max – 63

$S\to 61$ $\quad 9\,9\,9\,9\,9\,9\,7\to\frac{7!}{6!}$

$9\,9\,9\,9\,9\,8\,8\to\frac{7!}{5!2!}$

$S\to 60$ $\quad 9\,9\,9\,9\,9\,9\,6\to\frac{7!}{6!}=7$

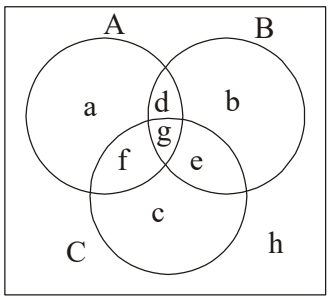
$9\,9\,9\,9\,9\,8\,7\to\frac{7!}{5!}=42$

$$9\,9\,9\,9\,8\,8\,8 \rightarrow \frac{7!}{4!3!} = 35$$

Q.7 H O N O L U L U Find total no. of 8 letter words that can be formed such that no two alike letters are together.

Ans 2220

Sol. OO LL UU HN



A ≡ O's are together

B ≡ L's are together

C ≡ U's are together

$$a+b+c+d+e+f+g+h = \frac{\lfloor 8}{\lfloor 2 \times \lfloor 2 \times 2} = 5040$$

g $\boxed{OO}$ $\boxed{LL}$ $\boxed{UU}$ HN

$$g = 15 \times \frac{\lfloor 2}{\lfloor 2} \times \frac{\lfloor 2}{\lfloor 2} \times \frac{\lfloor 2}{\lfloor 2} = 360$$

$d+g$ $\boxed{OO}$ $\boxed{LL}$ UU HN

$$d+g = \frac{\lfloor 6}{\lfloor 2} \times \frac{\lfloor 2}{\lfloor 2} \times \frac{\lfloor 2}{\lfloor 2}$$

$$d = 240 = e = f$$

$a+d+f+g$

$\boxed{OO}$ LL UU HN

$$a+d+f+g = \frac{\lfloor 7}{\lfloor 2 \lfloor 2} \times \frac{\lfloor 2}{\lfloor 2} = 1260$$

$$a = 660 = b = c$$

$$\Rightarrow h = 2220$$

Q.8 Four faces of a tetrahedral dice are marked with 2, 3, 4, 5. The lowest face being considered as the outcome. In how many ways a total of 30 can occur in 7 throws.

Ans: 413

Sol 5555532 ; 5555442 ; 5555433 ; 5554443 ; 5544444] [ ]

5.2 PERMUTATION OF ALIKE OBJECTS TAKEN SOME AT A TIME

Q.1 DADDY Find number of selection of 2 letters

Ans 4

Sol D D D A Y

Type	Example	selection
2	DD	${}^{1}C_{1}=1$
1+1	DA	${}^{3}C_{2}=3$
		4

Q.2 H O N O L U L U

(a) Find number of selection of 3 letters

Ans 22

Sol O O L L U U H N

Type	Example	selection
2H	OOL	${}^{3}C_{1}\,{}^{4}C_{1}=12$
1+1+1	OLU	${}^{5}C_{3}=10$
		22

(b) Find number of selection of 4 letters

Ans 26

Sol

Type	Example	selection
2+2	OOLL	${}^{3}C_{2}$
2+1+1	OOLU	${}^{3}C_{1}\,{}^{4}C_{2}=18$
1+1+1+1	OLUN	${}^{5}C_{4}=5$
		26

Q.3 M I S S I S S I P P I

(a) Find number of selection of 4 letters

Ans 21

Sol

Type	Example	selection
4	SSSS	${}^{2}C_{1}=2$
3+1	SSSP	${}^{2}C_{1}\times{}^{3}C_{1}=6$
2+2	SSII	${}^{3}C_{2}=3$
2+1+1	SSIP	${}^{3}C_{1}\times{}^{3}C_{2}=9$
1+1+1+1	SIPM	1
		21

(b) Find number of selection of 5 letters and number of 5 letter words

Ans 25,550

Sol

Type	Example	selection	5 letter words
$4+1$	SSSI	${}^{2}C_{1}\times{}^{3}C_{1}=6$	$\frac{\lfloor 5}{\lfloor 4}\times 6$
$3+2$	SSSII	${}^{2}C_{1}\times{}^{2}C_{1}=4$	$\frac{\lfloor 5}{\lfloor 3\lfloor 2}\times 4$
$3+1+1$	SSSIP	${}^{2}C_{1}\times{}^{3}C_{2}=6$	$\frac{\lfloor 5}{\lfloor 3}\times 6$
$2+2+1$	SSIIP	${}^{3}C_{2}\times{}^{2}C_{1}=6$	$\frac{\lfloor 5}{\lfloor 2\lfloor 2}\times 6$
$2+1+1+1$	SSIPM	${}^{3}C_{1}\times{}^{3}C_{3}=3$	$\frac{\lfloor 5}{\lfloor 2}\times 3$
$1+1+1+1+1$	–	$\frac{0}{25}$	

Q.4 Find number of 6 letter words that can be formed containing 3 vowels and 3 consonants using letters of word INTEGRATION

Ans ${}^{6}C_{3}\times(33)\times(42)$

Sol I I E A O T T N N G R

vowels

Type	Example	selection	3 letter words
$2+1$	IIE	${}^{1}C_{1}\times{}^{3}C_{1}=3$	$3\times\frac{\lfloor 3}{\lfloor 2}=9$
$1+1+1$	IAE	${}^{4}C_{3}=\frac{4}{7}$	$4\times\lfloor 3=\frac{24}{3}$

Consonants

Type	Example	selection	3 letter words
$2+1$	TTN	${}^{2}C_{1}\times{}^{3}C_{1}=6$	$6\times\frac{\lfloor 3}{\lfloor 2}=18$
$1+1+1$	TNG	${}^{4}C_{3}=4$	$4\times\lfloor 3=\frac{24}{42}$

- -

${}^{6}C_{3}\times(\text{vowel})\times(\text{consonants})$

$\Rightarrow{}^{6}C_{3}\times(33)\times(42)$

6.1 CIRCULAR PERMUTATION

The number of circular arrangements of n different things $= \frac{n!}{n} = (n-1)!$

Q. 1 Four different persons are to be seated around a circular table find total number of ways.

Ans. 3!

Sol. $(4-1)! = 3!$

Q. 2 Four different persons are to be seated around a circular table on numbered chairs. Find total number of ways.

Ans. 4!

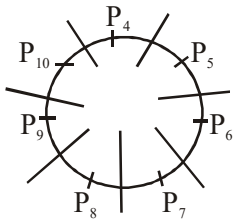
Sol. 4!

Q. 3 In how many ways $P_1, P_2, \ldots\ldots\ldots P_{10}$ are to be seated around a circle such that.
(i) If P_1, P_2 and P_3 are together. (ii) If P_1, P_2 and P_3 are separated.

Ans. (i) 7! 3! (ii) (6!) $({}^7C_3)$ (3!)

Sol. (i) $\boxed{P_1P_2P_3}\ P_4\ P_5\ P_6\ P_7\ P_8\ P_9\ P_{10} \rightarrow 7! \times 3!$

(ii)



$6! \times {}^7C_3 \times 3!$

Q. 4 7 Indians and 7 Americans are to be seated around a circular table such that
(i) No two Americans are together.
(ii)Such that Indians and Americans are alternate.

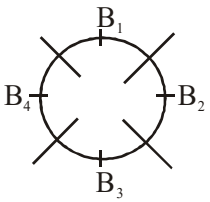
Ans. (i) (6!) $({}^7C_7)$(7!) (ii) (6!) $({}^7C_7)$ (7!)

Sol. 7I 7A
(i) $6! \times {}^7C_7 \times 7!$
(ii) same as part (i)

Q. 5 4 boys B_1, B_2, B_3, B_4 and 4 girls G_1, G_2, G_3, G_4 are to be seated around a circular table. Find total number of ways.
(ii) If all the girls are together. (iii) No two girls are together.
(iv) G_1 and G_2 are together. (v) B_1 and G_1 are not together.
(vi) Boys and girls are alternate.
(vii) Boys and girls are alternate and B_1 and G_1 are together.

Ans. (i) 7! (ii) 4! 4! (iii) 3! 4! (iv) 6! 2!
(v) (5!) $({}^6C_2)$ (2!) (vi) (3!) ${}^4C_4$ (4!) (vii) (2!) (3!) $({}^6C_1 \times 1)$

Sol. 4B 4G
(i) (8 – 1)!
(ii) $\boxed{G_1G_2G_3G_4}\ B_1\ B_2\ B_3\ B_4$
$(5-1)! \times 4!$

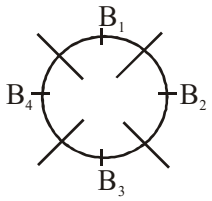
(iii) 

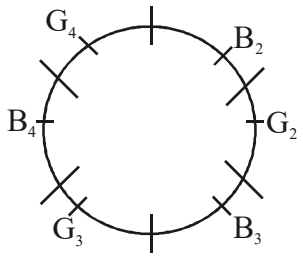
$(4-1)! \times {}^4C_4 \times 4!$

(iv) $G_1\ G_2\ G_3\ G_4\ B_1\ B_2\ B_3\ B_4$
$6! \times 2!$

(v) $7! - 6! \times 2!$ (total – ($B_1\ G_1$) together)

(vi) $(4-1)! \times {}^4C_4 \times 4!$



(vii) 

$(2! \times {}^3C_3 \times 3!) \times {}^6C_1 \times (1)$

Note

If clockwise & anti-clockwise circular permutations are considered to be same, then we will divide the answer by 2.

Q. 6 There are 10 different flowers available. In how many ways garlands consisting of 6 different flowers can be made?

Ans. $\dfrac{({}^{10}C_6)(5!)}{2}$

Sol. ${}^{10}C_6 \times \left(\dfrac{5!}{2}\right)$

Q. 7 In how many ways 10 persons can be seated around a circle such that not all persons have same neighbors in any two arrangements.

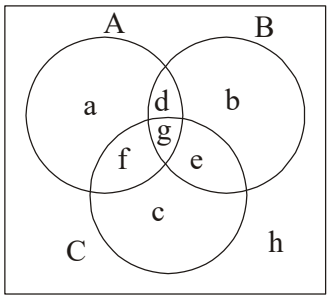
Ans. $\dfrac{9!}{2}$

Sol. $\dfrac{9!}{2}$

Q. 8 In how many ways all letters of word HONOLULU can be arranged around a circle such that no two alike letters are together.

Ans. 246

Sol.



A ≡ O's are together
B ≡ L's are together
C ≡ U's are together

$$a+b+c+d+e+f+g+h=\frac{7!}{2!2!2!}$$

$g \rightarrow \boxed{OO}\quad \boxed{LL}\quad \boxed{UU}\quad HN$

$g=(5-1)!=4!$

$d+g \rightarrow \boxed{OO}\quad \boxed{LL}\quad UU\quad HN$

$$d+g=\frac{(6-1)!}{2!}=\frac{5!}{2!}$$

$$d=\frac{5!}{2!}-4!=e=f$$

$a+d+f+g \rightarrow \boxed{OO}\quad LL\quad UU\quad HN$

$$a+d+f+g=\frac{(7-1)!}{2!2!}=\frac{6!}{2!2!}$$

$$a=\frac{6!}{2!2!}-d-f-g=b=c$$

HONOLULU

$$\text{Total}=\frac{7!}{2!2!2!}=\frac{7!}{8}$$

6.2 STATION METHOD IN CIRCULAR PERMUTATION

Q.1 A regular polygon of 15 sides is constructed. In how many ways can a triangle be formed using the vertices of the polygon such that no side of triangle is same as that of polygon ?

Ans 275

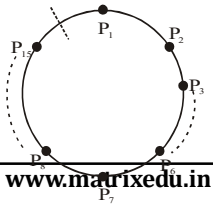
Sol. Select one point out of 15 point, therefore total number of ways = $^{15}C_1$

Suppose we select point P_1. Now we have to choose 2 more point which are not consecutive.

since we can not select P_2 & P_{15}.

Total points left are 12.

Now we have to select 2 points out of 12 points which are not consecutive



Total ways = $^{12-2+1}C_2 = {}^{11}C_2$

Every select triangle will be repeated 3 times.

So total number of ways = $\frac{^{15}C_1 \times {}^{11}C_2}{3} = 275$

Alternative :

First of all let us cut the polygon between points P_1 & P_{15}. Now there are 15 points on a straight line and we have to select 3 points out of these, such that the selected points are not consecutive.

x O y O z O w

Here bubbles represents the selected points,

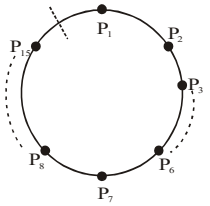
x represents the number of points before first selected point,

y represents the number of points between Ist & IInd selected point,

z represents the number of points between IInd & IIIrd selected point

and w represents the number of points after IIIrd selected point.

$x + y + z + w = 15 - 3 = 12$



here $x \geq 0, y \geq 1, z \geq 1, w \geq 0$

Put $y = 1 + y'$ & $z = 1 + z'$ $(y' \geq 0, z' \geq 0)$

$\Rightarrow x + y' + z' + w = 10$

Total number of ways = $^{13}C_3$

These selections include the cases when both the points P_1 & P_{15} are selected. We have to remove those cases.

Here a represents number of points between P_1 & 3rd selected point & b represents number of points between 3rd selected point and P_{15}

$\Rightarrow a + b = 15 - 3 = 12 \quad (a \geq 1, b \geq 1)$

put $a = 1 + t_1$ & $b = 1 + t_2$

$t_1 + t_2 = 10$

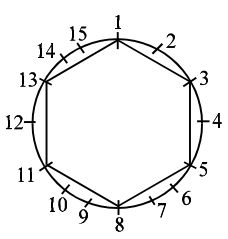
Total number of ways = $^{11}C_1 = 11$

Therefore required number of ways = $^{13}C_3 - {}^{11}C_1 = 286 - 11 = 275$

Q.2 How many hexagons can be constructed by joining the vertices of a quindecagon (15 sides) if none of the sides of the hexagon is also the side of the 15-gon.

Ans 140

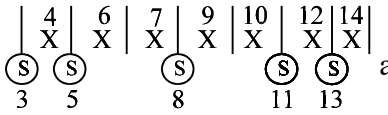
[Sol. Step (1) Select the initial vertex say '1' (in $^{15}C_1$ ways)



(2) Now 2 and 15 cannot be selected. From the remaining vertices 3 to 14 (twelve) we have to select 5 more for our hexagon.

(3) Symbolise the vertices to be taken by (S) (S) (S) (S) (S) and the vertices not be taken (7 in this case) by X X X X X X X.

(4) Identify the gaps between these crosses (8 in this case) and select



any five out of these gaps in $^8C_5$ ways.

(5) Serial number are to be alloted either to Ⓢ or to a 'X' whichever comes earlier. In the present problem the vertices corresponding to one selection are 3, 5, 8, 11, 13 and the hexagon as shown

(6) For each selection we therefore have a hexagon with two non consecutive vertices

(7) Number of hexagons = $^{15}C_1 \times {}^8C_5$. However this particular hexagon 1, 3, 5, 8, 11, 13 will occur 6 times when we select the initial vertex as 3 or 5 or 8 or 11 or 13. Hence our answer is 6 times more.

(8) Required number of hexagons = $\frac{15 \times {}^8C_5}{6} = \frac{15 \times 56}{6} = 140$ Ans]

[**Alternative Solution (Similar to that in linear) :**

As in linear, let us open this chain to have 1, 2, 3, ...13, 14, 15
O O O O O O to be selected ; | X | X | X | X | X | X | X | X | not to be selected
Number of ways = $^{10}C_6$
Required number of ways = $^{10}C_6$ – number of ways when 1 and 15 are included , since in circular these become consecutive
Now if 1 and 15 are already selected , 2 and 14 cannot be taken. Remaining vertices are
3, 4, 5, ... , 11, 12, 13 (11) ; O O O O (4) ; | X | X | X | X | X | X | (7)
Cases to be rejected = $^8C_4$
Required number of ways = $^{10}C_6 - {}^8C_4 \Rightarrow {}^{10}C_4 - {}^8C_4 \Rightarrow 210 - 70 = 140$]

7.1 TOTAL NUMBER OF COMBINATIONS

Q. 1 Find total number of combinations(selection of at least one thing) of four distinct letters A, B, C and D.

Ans. $2^4 - 1$

Sol. $2^4 - 1$

Q. 2 Find total number of combinations of A,A,A,B,B,B,C,C,D,D,E

(i) Selection of at least 1 letter (ii)Selection if at least 2A's are selected
(iii) At least one letter of each type (iv) At most 2A' s are selected.

Ans. (i) $4 \times 4 \times 3 \times 3 \times 2 - 1$ (ii) $2 \times 4 \times 3 \times 3 \times 2$
(iii) $3 \times 3 \times 2 \times 2 \times 1$ (iv) $3 \times 4 \times 3 \times 3 \times 2 - 1$

Sol. AAA BBB CC DDE
(i) $(3 + 1)(3 + 1)(2 + 1)(2 + 1)(1 + 1) - 1$
(ii) $2 \times (3 + 1)(2 + 1)(2 + 1)(1 + 1)$
(iii) $3 \times 3 \times 2 \times 2 \times 1$

(iv) $(2 + 1)(3 + 1)(2 + 1)(2 + 1)(1 + 1) - 1$

Q. 3 Four apples, five mangoes and 6 oranges are available. Find the number of combinations.

(i) Fruits of same species are alike.

(a) Selection of at least one fruit. (b)Selection of at least one fruit of each species.

(ii)Fruits of same species one different

(a) Selection of at least one fruit. (b) Selection of at least one fruit of each species.

Ans. (i) (a) 209 (b) 120

(ii) (a) $2^{15} - 1$ (b) $(2^4 - 1)(2^5 - 1)(2^6 - 1)$

Sol. 4A 5M 6O

(i) (a) $(4 + 1)(5 + 1)(6 + 1) - 1$

(b) $4 \times 5 \times 6$

(ii) (a) $2^{15} - 1$

(b) $({}^4C_1 + {}^4C_2 ..... {}^4C_4)({}^5C_1 + {}^5C_2 ..... {}^5C_5)({}^6C_1 + {}^6C_2 ..... {}^6C_6)$

$= (2^4 - 1)(2^5 - 1)(2^6 - 1)$

7.2 DIVISORS BASED PROBLEMS

Note :

(i) Every natural number except 1 has atleast 2 divisors. If it has exactly two divisors then it is called a prime. System of prime numbers begin with 2. All primes except 2 are odd.

(ii) A number having more than 2 divisors is called composite. 2 is the only even number which is not composite.

(iii) Two natural numbers are said to be relatively prime or coprime if their HCF is one. For two natural numbers to be relatively prime, it is not necessary that one or both should be prime. It is possible that they both are composite but still coprime, eg. 4 and 25.

(iv) 1 is neither prime nor composite however it is co-prime with every other natural number.

(v) Two prime numbers are said to be twin prime numbers if their non-negative difference is 2 (e.g.5 & 7, 19 & 17 etc).

(vi) All divisors except 1 and the number itself are called proper divisors.

Q.1 N = 75600 Find

(i) Total number of divisors

Ans 120

Sol $N = 75600 = 2^4 \cdot 3^3 \cdot 5^2 \cdot 7^1$

Total no. of divisors = 5.4.3.2 = 120

(ii) Total number of proper divisors

Ans 118

Sol = 120 – 2 (note that 1 and N are not the proper divisors)

(iii) Total number of even divisors

Ans. 96

Sol. Let divisor be of the type = $2^{P_1}3^{P_2}5^{P_3}7^{P_4}$

$P_1 \to 1$ to $4 \Rightarrow 4 \times (3+1)(2+1)(1+1)$

(iv) Total number of odd divisors

Ans 24

Sol no 2 should be taken.

Total Number= 4.3.2 = 24

(v) Total number of divisors divisible by 24

Ans. 36

Sol. $24 \to 2^3 \times 3^1$

$P_1 \to 3, 4$

$P_2 \to 1, 2, 3$

$= 2 \times 3 \times 3 \times 2$

(vi) Sum of all divisors

Ans $(2^0+2^1+2^2+2^3+2^4)(3^0+3^1+3^2+3^3)(5^0+5^1+5^2)(7^0+7^1)$

Sol. $S = (2^0 + 2^1 + 2^2 + 2^3 + 2^4)(3^0 + 3^1 + 3^2 + 3^3)$
$(5^0 + 5^1 + 5^2)(7^0 + 7^1)$

$$\left(\frac{2^5-1}{2-1}\right)\left(\frac{3^4-1}{3-1}\right)\left(\frac{5^3-1}{5-1}\right)\left(\frac{7^2-1}{7-1}\right)$$

(vii) Sum of all even divisors

Ans. $(2^1+2^2+2^3+2^4)(3^0+3^1+3^2+3^3)(5^0+5^1+5^2)(7^0+7^1)$

Sol. $S = (2^1 + 2^2 + 2^3 + 2^4)(3^1 .... 3^3)(5^1 + 5^2)(7^1)$

(viii) Sum of all divisors divisible by 24

Ans. $(2^3+2^4)(3^1+3^2+3^3)(5^0+5^1+5^2)(7^0+7^1)$

Sol. $S = (2^3 + 2^4)(3^1 + ... + 3^3)(5^0 + ... + 5^2)(7^0 + 7^1)$

(ix) Sum of all odd divisors.

Ans. $(2^0)(3^0+3^1+3^2+3^3)(5^0+5^1+5^2)(7^0+7^1)$

Sol. $S = 2^0(3^0 + ... + 3^3)\cdot(5^0 + ... 5^2)(7^0 + 7^1)$

(x) Number of ways in which N can be expressed as product of two divisors.

Ans. 60

Sol. $N = a \cdot b$

$N = 1 \cdot 75600$

$2 \cdot 37800$

$3 \cdot 25200$

$\vdots$

$75600 \cdot 1$

$$\text{No. of ways} = \frac{\text{No. of divisors}}{2} = \frac{120}{2} = 60$$

If no. of divisors are odd (n is a perfect square)

$$= \frac{\text{no. of divisors} + 1}{2}$$

(xi) Product of all the divisors

Ans. $(75600)^{60}$

Sol. $P = 1\cdot2\cdot3.....37800,75600$

$$P = \begin{pmatrix}1\\75600\end{pmatrix}\cdot\begin{pmatrix}2\\37800\end{pmatrix}\cdot\begin{pmatrix}3\\25200\end{pmatrix}.....\begin{pmatrix}25200\\3\end{pmatrix}\cdot\begin{pmatrix}37800\\2\end{pmatrix}\cdot\begin{pmatrix}75600\\1\end{pmatrix}$$

$P^2 = 75600\cdot75600.....\cdot75600$

$P^2 = (75600)^{120}$

$$P = (75600)^{\frac{120}{2}},\ P = (N)^{\frac{\text{No. of divisors}}{2}}$$

(xii) Number of ways in which N can be expressed as product of 2 divisors which are relatively prime.

Ans. 8

Sol. $N = 2^4\cdot3^3\cdot5^2\cdot7^1$

No. of ways $= 2^{(P-1)}$ where P = No. of primes

$= 2^{(4-1)} = 2^3 = 8$

8.1 MAXIMUM VALUE OF ${}^nC_r$

Q.1 Find value of r for which following binomial coeff. attains its maximum value.

(1) ${}^{40}C_r$ $r = 20$ (4) ${}^{37}C_r$ $r = 18 \text{ or } 19$

(2) ${}^{32}C_r$ $r = 16$ (5) ${}^{59}C_r$ $r = 29 \text{ or } 30$

(3) ${}^{56}C_r$ $r = 28$ (6) ${}^{41}C_r$ $r = 20 \text{ or } 21$

Sol. (i) ${}^{40}C_r \Rightarrow r = \frac{40}{2} = 20$

(ii) ${}^{32}C_r \Rightarrow r = \frac{32}{2} = 16$

(iii) ${}^{56}C_r \Rightarrow r = \frac{56}{2} = 28$

(iv) ${}^{37}C_r \Rightarrow r = \frac{37-1}{2}$ or $\frac{37+1}{2} \Rightarrow 18 \text{ or } 19$

(v) ${}^{59}C_r \Rightarrow r = \frac{59-1}{2} \text{or} \frac{59+1}{2} \Rightarrow 29 \text{ or } 30$

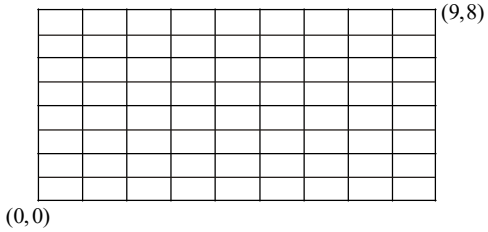
(vi) ${}^{41}C_r \rightarrow r = \frac{41-1}{2} \text{or} \frac{41+1}{2} \Rightarrow 20 \text{ or } 21$

8.2 GRID PROBLEM

Q.1 Find number of ways in which a person can travel from $(0,0)$ to $(9,8)$ along the grid via shortest path.

Ans $\frac{\lfloor 17}{\lfloor 9 \lfloor 8}$

Sol



9H 8V

H H H – – – H V V V – – – – V

9 8

Number of ways = $\frac{\lfloor 17}{\lfloor 9 \lfloor 8}$

Q.2 A (2, 1), B(7, 8), C(11, 12) Find number of ways for a person can travel along the grid via shortest path

(i) A to C (ii) A to C via B (iii) A to C but he cannot visit B

(iv) A to C if exactly one diagonal step is taken.

(v) A to C if exactly two diagonal steps are taken.

Ans. (i) $^{20}C_9$ (ii) $^{12}C_5\ ^8C_4$ (iii) $^{20}C_9 - {}^{12}C_5\ ^8C_4$ (iv) $\frac{19!}{8!10!1!}$ (v) $\frac{18!}{7!9!2!}$

Sol. A (2,1)
B (7, 8)
C (11, 12)

(i) $A \to C \to 9H11V \Rightarrow \frac{20!}{9!11!}s$

(ii) $A \to B \to C \Rightarrow (5H7V) \times (4H4V) = \frac{12!}{5!3!} \times \frac{8!}{4!4!}$

(iii) Total – via B = (i) – (ii)

(iv) 1 D 8 H 10 V $\Rightarrow \frac{19!}{8!10!}$

(v) 2D 7H 9V $\Rightarrow \frac{18!}{2!7!9!}$

8.3 DE-ARRANGEMENT

1. L_1 $D_1 = 0$

E_1

2. $L_1\ L_2$ Total $= 2C + 2W \Rightarrow 2! = 1 + D_2$

$E_1\ E_2$ $\Rightarrow$ $2 = 1 + D_2$ $\Rightarrow$ $D_2 = 1$

3. $L_1\ L_2\ L_3$ $\quad$ Total $= 3C + 1C2W + 3W \Rightarrow \underline{|3} = 1 + {}^3C_1 \times D_2 + D_3$

$E_1\ E_2\ E_3$ $\quad \Rightarrow \quad 6 = 1 + 3(1) + D_3 \Rightarrow \quad D_3 = 2$

4. $L_1\ L_2\ L_3\ L_4$ $\quad$ Total $= 4C + 2C2W + 1C3W + 4W \Rightarrow \underline{|4} = 1 + {}^4C_2 \times D_2 + {}^4C_1 \times D_3 + D_4$

$E_1\ E_2\ E_3\ E_4$ $\quad \Rightarrow \quad 24 = 1 + 6 + 8 + D_4 \Rightarrow \quad D_4 = 9$

5. $L_1\ L_2\ L_3\ L_4\ L_5$ $\quad$ Total $= 5C + 3C2W + 2C3W + 1C4W + 5W$

$D_1\ D_2\ D_3\ D_4\ D_5$ $\quad 120 = 1 + {}^5C_3 \times D_2 + {}^5C_2 \times D_3 + {}^5C_1 D_4 + D_5 \Rightarrow \quad D_5 = 120 - 76 = 44$

There are n letters and n corresponding envelopes. The number of ways in which letters can be placed in the envelopes (one letter in each envelope) so that no letter is placed in correct envelope is

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} + ... + \frac{(-1)^n}{n!}\right]$$

Q.1 Five letters are to arranged in 5 directed envelops. Find total number of ways if at least 3 letters are wrongly matched.

Ans. 109

Sol. 5W + 1R 4W + 2R 3W

$D_5 + {}^5C_1\ D_4 + {}^5C_2\ D_3$

$= 44 + 5 \times 9 + 10 \times 2$

$= 44 + 45 + 20$

$= 109$

8.4 SUMMATION OF NUMBERS

Q. 1 Find sum of all 5 digit numbers without repetition using 1, 2, 3, 4, 5.

Sol Sum due to 1

at unit's place	$----1$	$24 \times 1 \times 1$
10's	$---1-$	$24 \times 1 \times 10$
100's	$--1--$	$24 \times 1 \times 100$
1000's	$-1---$	$24 \times 1 \times 1000$
10000'a	$1----$	$24 \times 1 \times 10000$

Sum due to $1 = 24 \times 1 \times (11111)$

$2 = 24 \times 2 \times (11111)$

$3 = 24 \times 3 \times (11111)$

$4 = 24 \times 4 \times (11111)$

$5 = 24 \times 5 \times (11111)$

Sum $= 24 \times (1 + 2 + 3 + 4 + 5) \times (11111)$

$= \alpha \times (\text{sum of digits}) \times (11.....1)$

n times in case of n digit no.

$\alpha =$ number of times any digit is occuring at a particular place.

Q. 2 Find sum of all 5 digit numbers using 1, 2, 3, 4, 5 if repetition is allowed.

Ans. $625 \times (1 + 2 + 3 + 4 + 5)\ (11111)$

Sol. $\overline{5} \times \overline{5} \times \overline{5} \times \overline{5} \times \overline{\underline{1}}$

$f = 5^4$

$S = 5^4 \times (1 + 2 + 3 + 4 + 5) \times 11111$

Q. 3 Find sum of all four digit numbers without repetition using 1, 2, 3, 4, 5, 6.

Ans. $60 \times (1 + 2 + 3 + 4 + 5 + 6)\ (1111)$

Sol. 1,2,3,4,5,6

$_\ _\ _\ \underline{1}$

$f = 5 \times 4 \times 3 = 60$

$S = 60 \times (1 + 2 + 3 + 4 + 5 + 6) \times 1111$

Q. 4 Find sum of all 5 digit number using 0, 1, 2, 3, 4, 5 without repetition.

Ans. $(120 \times (0 + 1 + 2 + 3 + 4 + 5)\ (11111)) - (24\ (1 + 2 + 3 + 4 + 5)\ (1111))$

Sol. 0,1,2,3,4,5

Total $\Rightarrow$ $_\ _\ _\ _\ \underline{0}$

$f_1 = 5 \times 4 \times 3 \times 2 = 120$

$S_1 = 120 \times (0 + 1 + 2 + 3 + 4 + 5) \times 11111$

$S_1 = 120 \times 15 \times 11111$

Starting with zero

$f_2 = 4 \times 3 \times 2 = 24$

$S_2 = 24 \times (1 + 2 + 3 + 4 + 5) \times 11111$

$S = S_1 - S_2$

$S_2 = 120 \times 15 \times 11111 - 24 \times 15 \times 1111$

Q. 5 Find sum of all 6 digit numbers using 2, 3, 3, 4, 4, 4.

Ans. $(10 \times 2 \times 111111) + (20 \times 3 \times 111111) + (30 \times 4 \times (111111))$

Sol. 2,3,3,4,4,4

$\underline{2}\ _\ _\ _\ _\ _ \Rightarrow f_1 = \dfrac{5!}{2!3!} = 10$

$\underline{3}\ _\ _\ _\ _\ _ \Rightarrow f_2 = \dfrac{5!}{3!} = 20$

$\underline{4}\ _\ _\ _\ _\ _ \Rightarrow f_3 = \dfrac{5!}{2!2!} = 30$

Sum due to $2 = 10 \times 2 \times 111111$

Sum due to $3 = 20 \times 3 \times 111111$

Sum due to $4 = 30 \times 4 \times 111111$

9.1 DISTRIBUTION OF ALIKE OBJECTS

Number of ways in which n identical things may be distributed among p persons if each person may receive none, one or more things is ${}^{n+p-1}C_n$.

Q. 1 30 alike coins are to be distributed among 5 persons A, B, C, D and E. Find number of ways

(ii) Such that each of them gets at least one coin. (iii) Each of them gets at least two coins.

(iv) Such that A gets at least 3, B gets at least 2, C gets at least 3.

Ans. (i) $^{34}C_4$ (ii) $^{29}C_4$ (iii) $^{24}C_4$ (iv) $^{26}C_4$

Sol. $30 \rightarrow 5$

(i) $^{30+5-1}C_{5-1} = {}^{34}C_4$

(ii) $30 \rightarrow 5$, $25 \rightarrow {}^{25+5-1}C_{5-1} = {}^{29}C_4$

(iii) $30 \rightarrow 2+2+2+2+2$, $20 \rightarrow {}^{20+5-1}C_{5-1} = {}^{24}C_4$

(iv) $30 \rightarrow 3+2+3$, $22 \rightarrow {}^{22+5-1}C_{5-1} = {}^{26}C_4$

Q. 2 In how many ways 36 marks be allotted to 8 questions if atleast 2 marks are to be given to each question, assuming the marks are to be alloted in non negative integral values.

Ans. $^{27}C_7$

Sol. $36 \rightarrow 2\times8 = 16$, $20 \rightarrow {}^{20+8-1}C_{8-1} = {}^{27}C_7$

9.2 ALGEBRA BASED PROBLEMS (ON DISTRIBUTION OF ALIKE OBJECTS)

Q. 1 Find number of integral solutions of equation $x + y + z = 100$.

(i) Non-negative (ii)Natural solutions

(iii) $x \geq 3, y \geq 2, z \geq 1$ (iv) $x \geq 2, y > 1, z \geq 2$

(v) $x \geq 3, y \geq -3, z \geq 1$

Ans. (i) $^{102}C_2$ (ii) $^{99}C_2$ (iii) $^{96}C_2$ (iv) $^{96}C_2$ (v) $^{101}C_2$

Sol. (i) No. of solution $= 1^{00+3-1}C_{3-1}$

(ii) $100 \rightarrow 3 = 1+1+1$, $97 \rightarrow {}^{97+3-1}C_{3-1}$

(iii) $x \geq 3, y \geq 2, z \geq 1$

$100 \rightarrow 3+2+1$, $94 \rightarrow {}^{94+3-1}C_{3-1}$

(iv) $x \geq 2, y > 1, z \geq 2$

$x \geq 2, y \geq 2, z \geq 2$

(v) $x \geq 3, y \geq -3, z \geq 1$

$100 \rightarrow 3-3+1$, $99 \rightarrow {}^{99+3-1}C_{3-1}$

Q. 2 Find number of integral solutions of the following inequality $x + y + z + w \leq 30$

(i) Non-Negative (ii) Natural solutions

Ans. (i) $^{34}C_4$ (ii) $^{30}C_4$

Sol. $x + y + z + w \leq 30$

(i) $x + y + z + w + p$ (false begger) $= 30$

$^{30+5-1}C_{5-1}$

(ii) 30 → 4, 26

$x + y + z + w \le 26$

$x + y + z + w + \phi = 26$

$^{26+5-1}C_{5-1}$

Q. 3 Find number of non negative integral solutions of the inequality $x + y + z + w < 30$

Ans. $^{33}C_4$

Sol. $x + y + z + w < 30$

$x + y + z + w \le 29$

$x + y + z + w + p = 29$

$^{29+5-1}C_{5-1}$

Q. 4 Find number of non negative integral solutions of the inequality $20 \le x + y + z + w \le 30$

Ans. $^{34}C_4 - {}^{23}C_4$

Sol. $20 \le x + y + z + w \le 30$

$(x + y + z + w \le 19)$

$(x + y + z + w + p = 30) - (x + y + z + w + p = 19)$

$^{30+5-1}C_{5-1} - {}^{19+5-1}C_{5-1}$

Q. 5 Find number of dissimilar terms in the expansion of $(x_1+x_2+.....+x_n)^m$

Ans $^{m+n-1}C_m$

Sol $= \underbrace{00....0}_{m} \ \underbrace{\emptyset\emptyset....\emptyset}_{n-1} = {}^{m+n-1}C_m$

Alternate :

$(x_1 + x_2 + ... + x_n)^m$

General term $= kx_1^{P_1} \cdot x_2^{P_2} .....x_n^{P_n}$

$P_1 + P_2 + P_3 + ... P_n = m$

$= {}^{m+n-1}C_{n-1}$

Q. 6 In an ice-cream shop there are 10 different varieties of icecream. Ice-cream of same varieties are all alike. In how many ways 4 ice-creams can be selected assuming there are sufficient number of ice-creams of each variety.

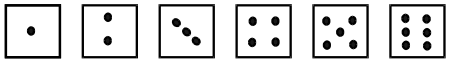
Ans. $^{13}C_9$

Sol. $x_1 + x_2 + ... x_{10} = 4$

$^{4+10-1}C_{10-1}$

Q. 7 n alike dice are thrown. How many different results can be obtained.

Ans $^{n+5}C_n$

Sol 

treat these six faces as beggars.

$\underbrace{000....0}_{n \text{ throws}} \ \underbrace{\emptyset\emptyset...\emptyset}_{5} = \dfrac{(n+5)!}{n!\ 5!} = {}^{n+5}C_5 = {}^{n+5}C_n$

Alternate :

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = n$

$^{n+6-1}C_{6-1}$

Q. 8 There are unlimited number of red, white, blue and green balls. Balls of same color are all alike. In how many ways 10 balls can be selected.

Ans. $^{13}C_3$

Sol. $x_R + x_W + x_B + x_y = 10$

$^{10+4-1}C_{4-1}$

Q.9 Number of points with integral co-ordinates that lie inside a triangle whose co-ordinates are (0, 0), (0, 102) and (102,0) is

Ans : 5050

Sol [Hint : $x + y < 102 \quad x \in N, \ y \in N$

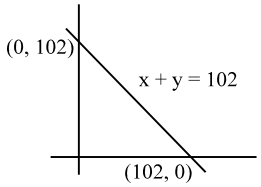
i.e. $x + y \le 101$

give 1 to each x and y

hence $x + y \le 99$

equivalent to $x + y + z = 99$

$\Rightarrow \quad ^{101}C_2 = \dfrac{101.100}{2} \quad \underbrace{0000....00}_{99}\ \underbrace{\emptyset\emptyset}_{2} = 5050$



9.3 COEFFICIENT METHOD

Q.1 Find number of non-negative integral solutions of the following equations

(i) $x + y = 5, x \le 4, y \le 5$

(ii) $x + y + z = 100, x \le 70, y \le 70, z \le 100$

Q.2 \\192.168.8.99\d\05-SESSION-2019-20\Question Bank JEE-main ONLINE _Jan. & April attempt_print\15_Jan. attempt_2025\Mathematics\Maths QB_FCopy Jan Attepmt 2025\Page46\Q65